

Arika Khor

SWE & Researcher with Interest in Autonomous Systems, Digital Facial Privacy Protection, and System Design

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Education

Expected B.S. 2026 , M.S. 2027	The University of Oklahoma B.S. / M.S. Computer Science, focus in Robotics & Computer Vision
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Experience

June 2026 - Current	The Boeing Company <i>Intern: Simulation Software Engineering</i> · Worked with podman containers for on-device builds of previous on-prem server only codebases. · Developed a hybrid Retrieval-Augmented Generation and Spec-Driven framework which: – Ran custom Sql lite database queries and creation per project item file. – Delegated subagents to speed up initial tasks such as codebase mapping and PDF reading. – Leveraged aforementioned framework to speed up accurate Software Design Document creation · Debugged CI/CD container pipelines for deployment to K8 clusters · Developed knowledge graph whcih
June 2025 - Current	Construction Innovation and Analytics Lab (CInnaLAB) <i>Time to Incident Predictor on a Autonomous vehicle in dynamic construction sites</i> · Implemented Autonomous ground robot navigating on known dynamic construction site. · Used: LIDAR, 3D AI stereo depth camera, and GNSS for localization and navigation. · Developed System Design for Time to Incident Predictor in a Construction Sites
Aug 2025 - Current	Autonomous Interactive Robotics Lab (AIROU) <i>Collaborative Autonomous Vehicles at Smart Intersections with Nvida Isaac Lab</i> · Developed Multi-agent camera-only reinforcement learning (RL) simulation vehicles at a smart intersection with Vehicle-to-Infrastructure communication. · Developed a self healing AI testing agent which would correct the RL policy without needing user intervention. <i>ARCPPro Autonomous Model Race Cars</i> · Developed Ansible images automating setup for each robot. · Created various front-end websites and webapps for robot UI and documentation. · Designed autonomous software stack with sensor fusion for cars which then drove around using Simultaneous Localization and Mapping (SLAM).
Sept 2022 - July 2024	Singapore Armed Forces <i>Curriculum Development Data Analyst</i> · Used python & pandas for Prediction based data analysis on Cadet scores using pandas, simple linear regression, and · Migrated excel workbooks used in trainee scoring to more modern processing with python (todo)
Jan 2020 - Dec 2020	GameDuchy Game Development Company <i>Assistant Community Relations Manager</i> · Implemented custom chat bot which fetched data from a collection of wiki pages and outputted it in a human readable card. · Planned, organised and executed regular in-game community wide events, working with various departments to successfully implement these events.

Notable Personal Projects

2025 Hacklahoma	Real Time Local Facial Recognition · Ability to match or make new faces in real time on a live camera with mongodb in an isolated local system. · Web UI built with Streamlit and flask to merge/delete/assign/list saved faces.
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School Based organizations

Aug 2024 - Current	Sooner Competitive Robotics <i>Software Member & Treasurer (25-26)</i> · Basic PCB design with Altium for LED controller and CAD design with Onshape for small onboarding robot. · Developed real time mock log playback of our UI, running previous logs of robot in UI. · Collaborated in a large team using ROS2 pub-sub codebases and implemented robotic systems design.
Aug 2016 - May 2022	Scouting America <i>Various Leadership Positions</i> · Eagle Scout Project: Designed and built several weather radio antennas by organizing a large group to mass-produce them. They were then given to the city's emergency coordinator to distribute to public service buildings.
Aug 2024 - Current	Sooner Competitive Programming Team <i>Team Capitan (24-26, Team Advisor (26-27)</i>

Technical Skills

Core Competencies

- Systems Design, Autonomous Systems, Program Coordination, Cross-Functional Teamwork, Technical Documentation, Agile/Scrum.

Tools & Infrastructure

- Docker, Podman, Kubernetes, Ansible, Git, Jira, Jenkins, Tailscale, Linux Ubuntu, Debian.

Programming Languages

- Java (Spring, Thymeleaf), C++, C#, JavaScript, Python, Bash.

Frameworks & Databases

- ROS2, Postgres, MongoDB